

Isabelle Rose

Isabelle A. Rose

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Education

Oxford University - Master of Mathematical and Theoretical Physics 2:1

October 2025 - Present

- Classes in fundamental physics disciplines: quantum field theory, quantum information, quantum error correction, machine learning.
- Dissertation in Gauge Theory Gravity

Massachusetts Institute of Technology - Bachelor in Physics 4.7/5.0

September 2020 - May 2024

- Classes in fundamental physics: quantum mechanics, differential equations, fundamentals of programming
- Undergraduate research in quantum pilot wave hydrodynamics and Bell violations

Employment

Seal - AI & Algorithmic Process Automation

June 2026 - Present

- Engineered bespoke AI-integrated software architectures at client sites, translating user requests to automate data entry, sharing, and storage to systematically reduce persistent errors.
- Executed on-site system installations and led technical training programs, ensuring effective client adoption and utilization of the platform.

Lexington Science Academy - Physics Instructor

September 2024 - April 2026

- Managed client relationships and marketing, gaining seven new clients in the first three months of the company.
- Designed syllabi that taught university-level mathematics to high school students, honing my ability to communicate technical information to an audience of any level of familiarity.

MIT School of Mathematics - Undergraduate Researcher

October 2023 - June 2024

- Wrote Python scripts to simulate droplet trajectories through different environments, which used Monte Carlo and Diffusion process simulations.
- Analysed experimental data using pandas in python to support the possibility of classical Bell violation.
- Contributed to Professor Bush's next publication by providing a mechanism to achieve Bell violation through the use of integrals of non-commuting random variables. DOI: 10.1103/PhysRevFluids.9.084001

Skills

- Quantum Error Correction and Machine Learning
- Leetcode data structures and algorithms
- Python with numpy, sympy and pandas
- Natively fluent in English and French